



Decision Support System for Liver Trauma Assessment

Ph.D Pre-Proposal Presentation

by

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Presentation Overview

- Introduction
- Literature Review
- Research Gaps
- Objectives
- Research Gaps to Objectives Mapping
- Methodology
- Course Work Details
- Time Plan
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Introduction-I

- Liver Trauma [1]

- Injury to liver due to external forces
- ~5% of all trauma admissions
- Primary cause of death in severe abdominal trauma, 10-15% mortality rate

- Mechanism of injury [2,3]

- Most injured organ in blunt trauma and in penetrating trauma
- Blunt trauma
 - 35-45% of abdominal trauma
 - MVA, pedestrian accidents, falls, sports, fights
- Penetrating trauma
 - 40% SWs and 30% GSWs
 - Stabbing, gunshot, shear forces
- Right lobe of the liver most affected

Introduction-II

- Why liver is prone to trauma? [4,5,6]

Largest peritoneal abdominal organ in the human body

■ Location and Size:

- Situated in the RUQ
- ~2% total body mass
- Fixed position (falciform, coronary, triangular ligaments)

■ Fragility:

- Covered by thin fibrous capsule (Glisson capsule)
- Fragile parenchyma

■ Vascularity:

- Dual blood supply
 - hepatic artery (25%), portal vein (75%)
 - High blood flow, haemorrhage
- Biliary tree disruption

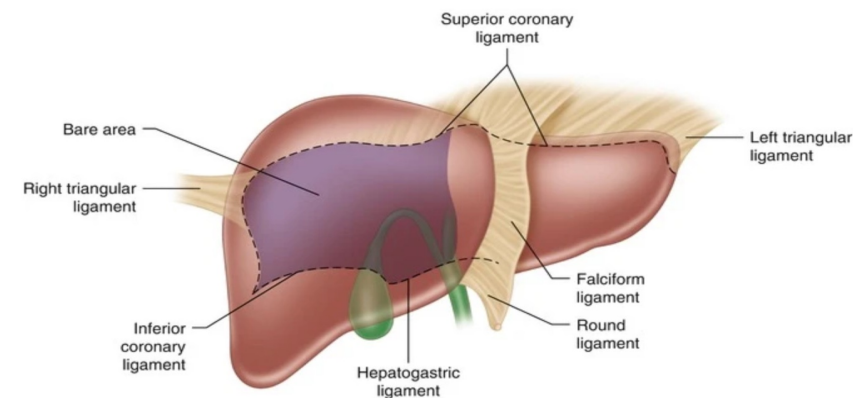


Figure 1.1: Liver Anatomy.

(Source:https://link.springer.com/chapter/10.1007/978-1-4939-1200-1_1)

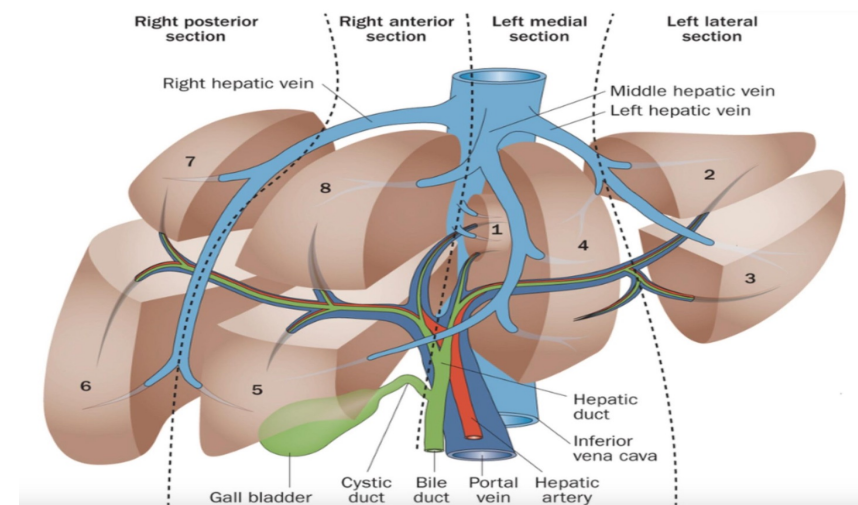


Figure 1.2: Liver segmented along Canaud

(Source:<https://ilts.org/insights/couinauds-segmental-anatomy-of-the-liver-and-proposed-classification-of-pediatric-liver-tumors/>)

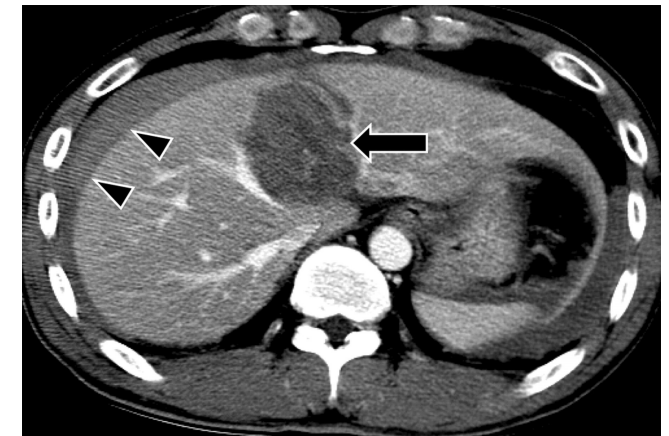
Introduction-III: Liver Trauma Injuries



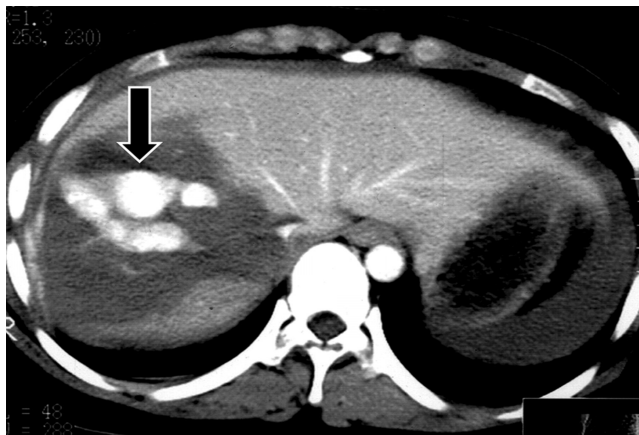
(a) Lacerations (arrow).



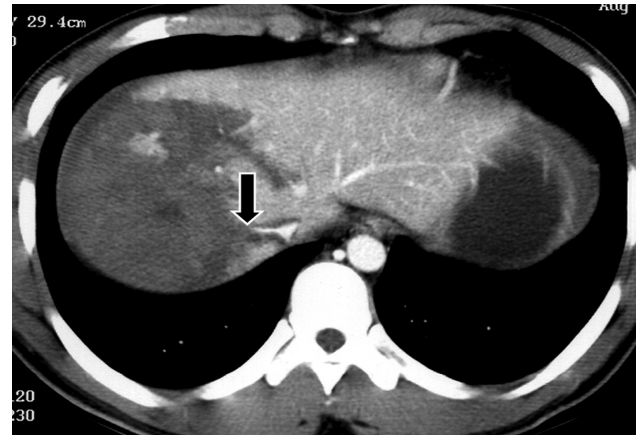
(b) Subcapsular Hematoma (arrow).



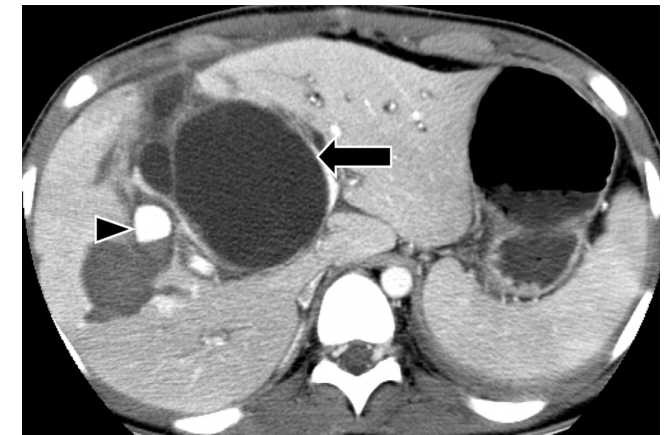
(c) Intraparenchymal Hematoma (arrow),
Subcapsular Hematoma (arrowhead).



(d) Hepatic vein injury: Active bleeding within
intraparenchymal hematoma.



(e) Hepatic venous injury: Lacerations extending
to Inferior Vena Cava (IVC) and cutoff of right
hepatic venous drainage (arrow).



(f) Biliary duct injury: Biloma (arrow),
pseudoaneurysm (arrowhead).

Figure 1.3: Different Liver Injuries Types (a), (b), (c), (d), (e), (f) [7].

Introduction-IV: AAST-Liver Injury Classification

AAST (American Association for the Surgery of Trauma) Liver Injury Scale (2018 revision) [8]

AAST Grade	Imaging Criteria (CT Findings)
I	Subcapsular hematoma <10% surface area Parenchymal laceration <1 cm depth
II	Subcapsular hematoma 10-50% surface area; intraparenchymal hematoma <10 cm in diameter Laceration 1-3 cm in depth and ≤10 cm length
III	Subcapsular hematoma >50% surface area; ruptured subcapsular or parenchymal hematoma Intraparenchymal laceration >10 cm Laceration >3 cm depth Any injury in the presence of a liver vascular injury or active bleeding contained within liver parenchyma
IV	Parenchymal disruption involving 25-75% of a hepatic lobe Active bleeding extending beyond the liver parenchyma into the peritoneum
V	Parenchymal disruption >75% of hepatic lobe Juxtahepatic venous injury to include retrohepatic vena cava and central major hepatic veins

Table 1.1: CT based AAST Liver Trauma Grading [8].

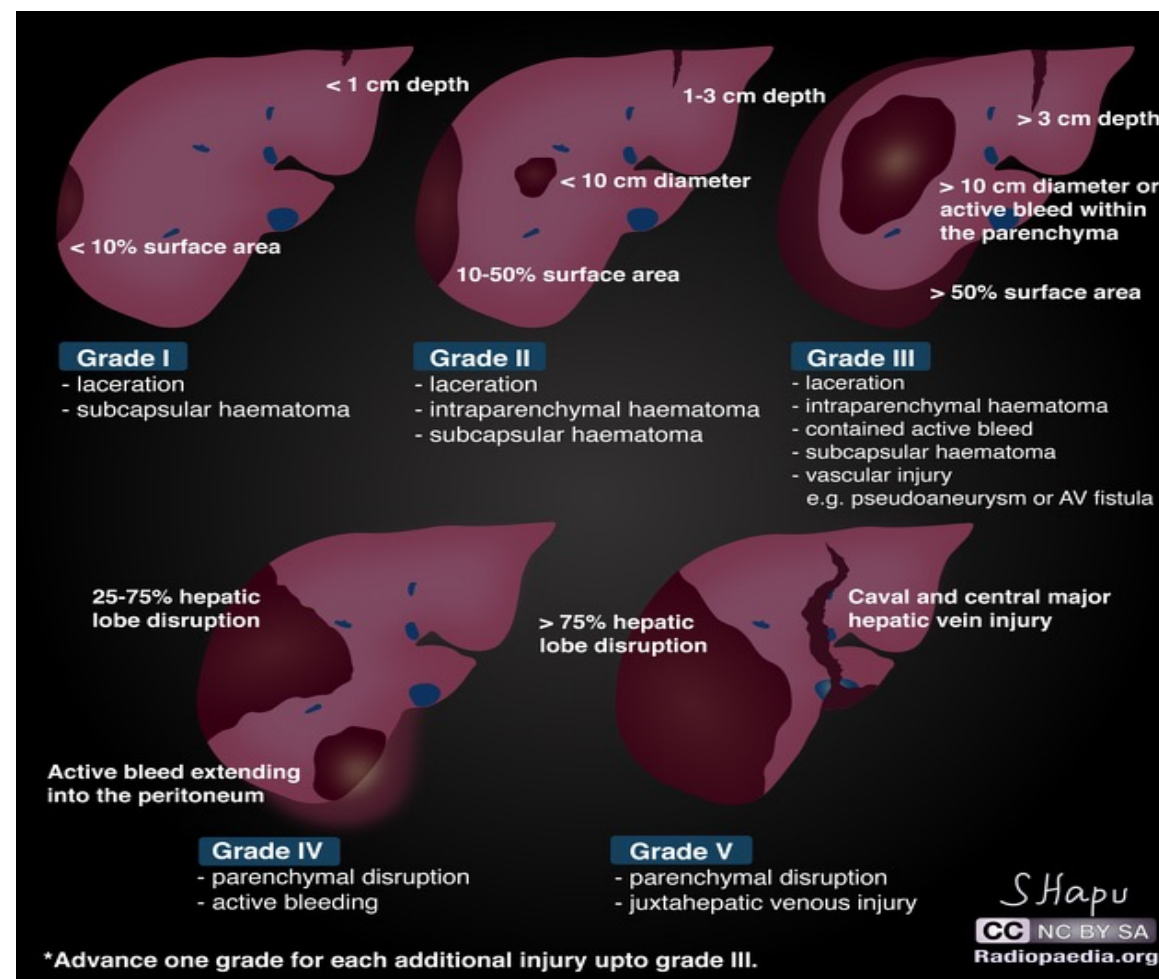


Figure 1.4: Liver Injury Grading as per AAST-OIS [9].

Introduction-V: WSES Trauma Classification

WSES (World Society of Emergency Surgery) [10]

- Liver trauma classification is based on AAST-OIS anatomic classification + hemodynamic status + associated injuries

	WSES grade	AAST	Hemodynamic	CT scan	First-line treatment
Minor	WSES grade I	I–II	Stable	Yes	NOM (Clinical/Laboratory/Radiological Evaluation)
Moderate	WSES grade II	III	Stable		
Severe	WSES grade III	IV–V	Stable		
	WSES grade IV	I–V	Unstable	No	OM

Table 1.2: WSES classification of Liver Trauma

Introduction-VI: Initial Assessment (Imaging)

- **Contrast-enhanced Computed Tomography (CT) [11]**
 - Gold standard: Sens./Spec.: 98%
 - Grade of injury
 - Active bleeding
 - Associated injuries
- **Why automated assessment? [12]**
 - CT limitations
 - Time consuming
 - Radiologist variability
 - Automated tools can:
 - Provide rapid, consistent, and accurate evaluations

Literature Review I

Author, Title, Journal	Dataset	Work-done	Findings
Farzaneh et al (2022) [13] <i>A Deep Learning Framework for Automated Detection and Quantitative Assessment of Liver Trauma.</i> BMC Medical Imaging	- University of Michigan Health System (UMHS) Department of Radiology (private: train + test) 3DIRCAD (public: test)	2 U-Net models used to segment liver parenchyma and trauma regions - Post processing done to refine the segmented trauma region by incorporating human domain knowledge and vision algorithms (morphological operations + 3D Chan-Vese active contour model (ACM)) to avoid false positives - Calculated the liver parenchyma disruption index (LDI) to assess the presence and extent of liver trauma	- The framework was trained on UMHS dataset and additional testing of liver parenchyma segmentation was done on 3DIRCAD dataset. - Achieved model performance for Di, Re, Pr = 96.13%, 96.00%, 96.35% for liver and 51.21% (>5%, 72.45%), 53.20% (>5%, 73.84%), 56.76% (>5%, 75.35%) for trauma segmentation. - Bland–Altman analysis was also performed to understand the algorithms agreement with the systematic errors and the ground truth. The model showed negligible bias (-0.13) - Performance of the liver trauma segmentation improved for considerable injuries with over 5% of LDI - The study is limited to liver intra-parenchymal hematoma.

Literature Review II

Author, Title, Journal	Dataset	Work-done	Findings
Huang et al (2022) [14] <i>Automated Quantitative Assessment of Pediatric Blunt Hepatic Trauma by Deep Learning-Based CT Volumetry</i> European Journal of Medical Research	CT images of 170 children for blunt liver trauma from Children's Hospital of Soochow University (private dataset)	▪ Leveraged 2 MONAI dynUNet models to segment liver parenchyma and trauma regions ▪ Histogram equalization and grayscale inversion was performed as pre-processing step on the output of 1st dynUNet model to highlight the trauma region before passing it to 2nd dynUNet model. ▪ Calculated the liver parenchyma disruption index (LDI) to assess the presence and extent of liver trauma	▪ The framework was trained on 118 CT images and validated on 52. ▪ Achieved model performance for Di, Re, Pr = 94.75%, 95.46%, 96.35% for liver and 72.91%, 76.8 %, 72.4 % for trauma segmentation. ▪ The Spearman rank correlation analysis was performed to show the correlation between automated LPDI and liver trauma volume with the AAST grade ($\rho=0.823$, $p<0.001$; $\rho=0.831$, $p<0.001$, respectively) ▪ The study is limited to liver intra-parenchymal hematoma. ▪ Performance degrades for lower AAST grade trauma regions.

Literature Review III

Author, Title, Journal	Dataset	Work-done	Findings
Dreizin et al (2021) [15] <i>Added Value of Deep Learning-Based Liver Parenchymal CT Volumetry for Predicting Major Arterial Injury after Blunt Hepatic Trauma.</i> Abdominal Radiology	Dataset of 73 adult patients who underwent contrast-enhanced admission CT of the abdomen followed by hepatic arterial angiography at two level I trauma centers (56 patients from R Adams Cow- ley Shock Trauma Center (STC) , and 17 patients from Ryder Trauma Center (RTC)) (private)	▪ Extensive voxel-wise ground-truth labeling was done. Volumetric measurements such as total liver parenchymal disruption (LPD) volume, total liver volume, subcapsular hematoma volume, and contrast extravasation (CE) volume was extracted from CT images ▪ Employed pre-trained DL models for whole organ and injury segmentation. 4-fold validation was used for training ▪ LDPI was calculated for liver trauma severity assessment. ▪ Two decision tree models created using the classification and regression tree (CART) approach: DR1: Combined automated LPDI and binary CE (CE as either present or absent). DR2: Combined automated LPDI and volumetric CE measurements (from manual segmentation).	▪ The automated results showed excellent correlation and concurrence with the manual measurements, with high ICC and Pearson's r values, for laceration vol, whole liver vol and LDPI. ▪ Bland-Altman plot showed minimal bias for LDPI. ▪ Accuracy, sensitivity, specificity, PPV and NPV was calculated for Binary CE, Decision Tree analysis, and AAST grade. ▪ DTA with auto LDPI + binary resulted in 32% reduction in false negative ▪ DTA with auto LDPI + CE volume resulted in 63% reduction in false negative ▪ Study only considered the CT scans of upper abdomen during portal venous phase ignoring the blood pooling in the lower abdomen. ▪ Also didn't consider injury near the bile duct and periphery.

Literature Review IV

Author, Title, Journal	Dataset	Work-done	Findings
Shen et al (2024) [16] <i>The Application of Deep Learning in Abdominal Trauma Diagnosis by CT Imaging.</i> World Journal of Emergency Surgery	▪ RSNA (Radiological Society of North America) dataset containing 3147 CT scans with various abdominal injuries from 23 sites and 14 countries. ▪ Dataset was categorized into six sections: liver, spleen, kidney, intestine, extravasation, and any injury. Each section divided into health or injury categories.	▪ Images cropped to focus on the abdominal organs within each scan. ▪ Designed a 2-stage model to detect trauma regions. ▪ Stage1-Segmentation (2D): Used U-Net model with EfficientNetB0 as backbone to segment individual organs ▪ Stage2: Injury Classification (2.5D): An EfficientNetB1 model was used to process the image from segmented slice. The LSTM (long short-term memory) network was introduced to get the spatial information between the segmented images. A Neck structure and Multi-Label classification was used to compute the average injury probability for each organ ▪ Used Grad-CAM to visualize model working.	▪ Classification metrics like accuracy, sensitivity and specificity was calculated ▪ Model performance was evaluated using ROC and AUROC ▪ The model obtained an accuracy of 0.873 (PPV: 0.789; NPV: 0.895; Sensitivity: 0.789; Specificity: 0.895; AUC:0.817) ▪ Overall, the model performance was good in diagnosing liver injuries ▪ The model is limited to only classification task – injured, healthy

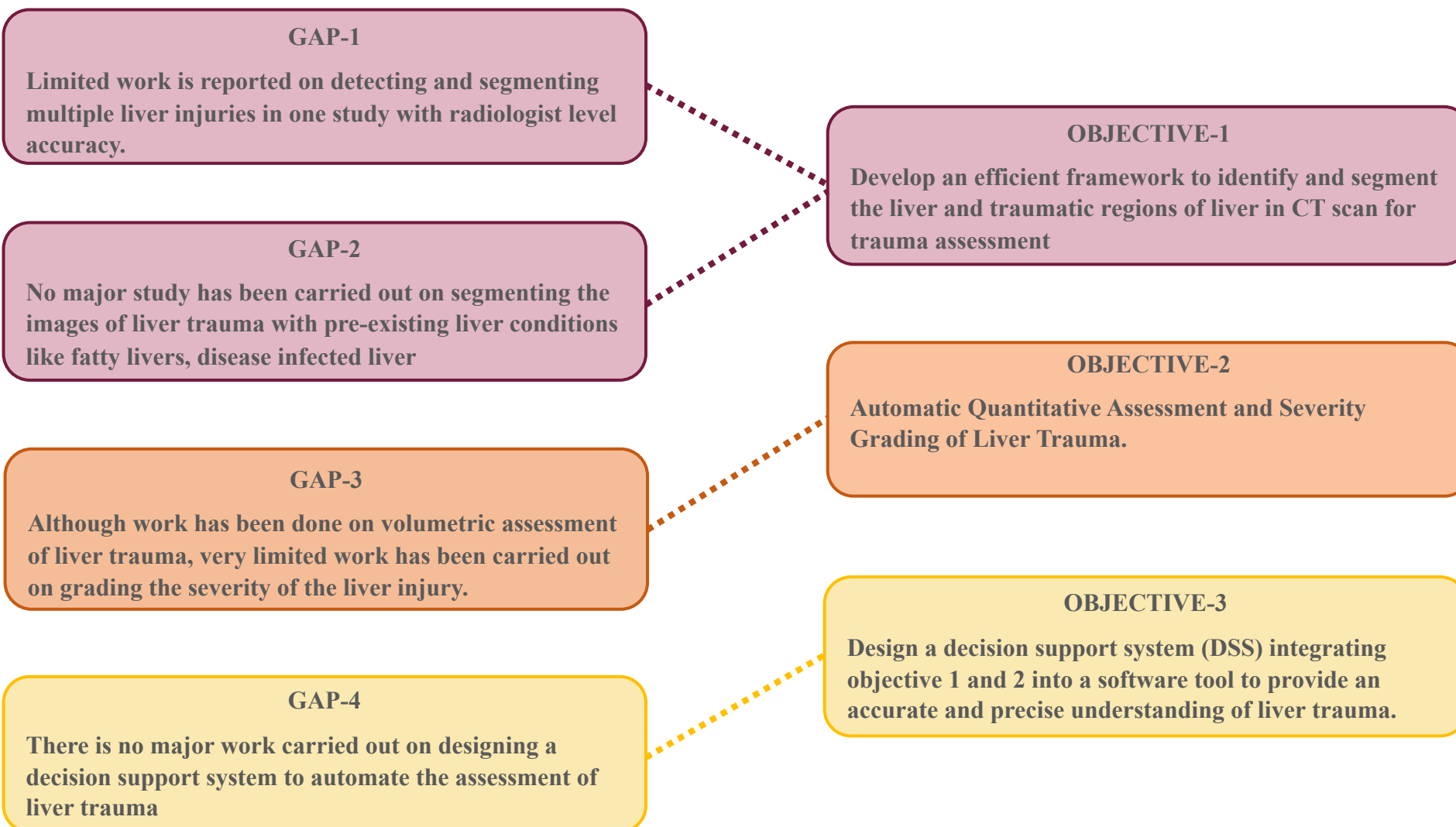
Research Gaps

1. Limited work is reported on detecting and segmenting multiple liver injuries in one study with radiologist level accuracy [13,15].
2. No major study has been carried out on segmenting the images of liver trauma with pre-existing liver conditions like fatty livers, disease infected liver etc [13].
3. Although work has been done on volumetric assessment of liver trauma, very limited work has been carried out on grading the severity of the liver injury [14].
4. There is no major work carried out on designing a decision support system to automate the assessment of liver trauma

Objectives

1. Develop an efficient framework to identify and segment the liver and traumatic regions of liver in CT scan for trauma assessment.
2. Automatic Quantitative Assessment and Severity Grading of Liver Trauma.
3. Design a decision support system (DSS) integrating objective 1 and 2 into a software tool to provide an accurate and precise understanding of liver trauma.

Research Gaps to Objectives Mapping



Methodology

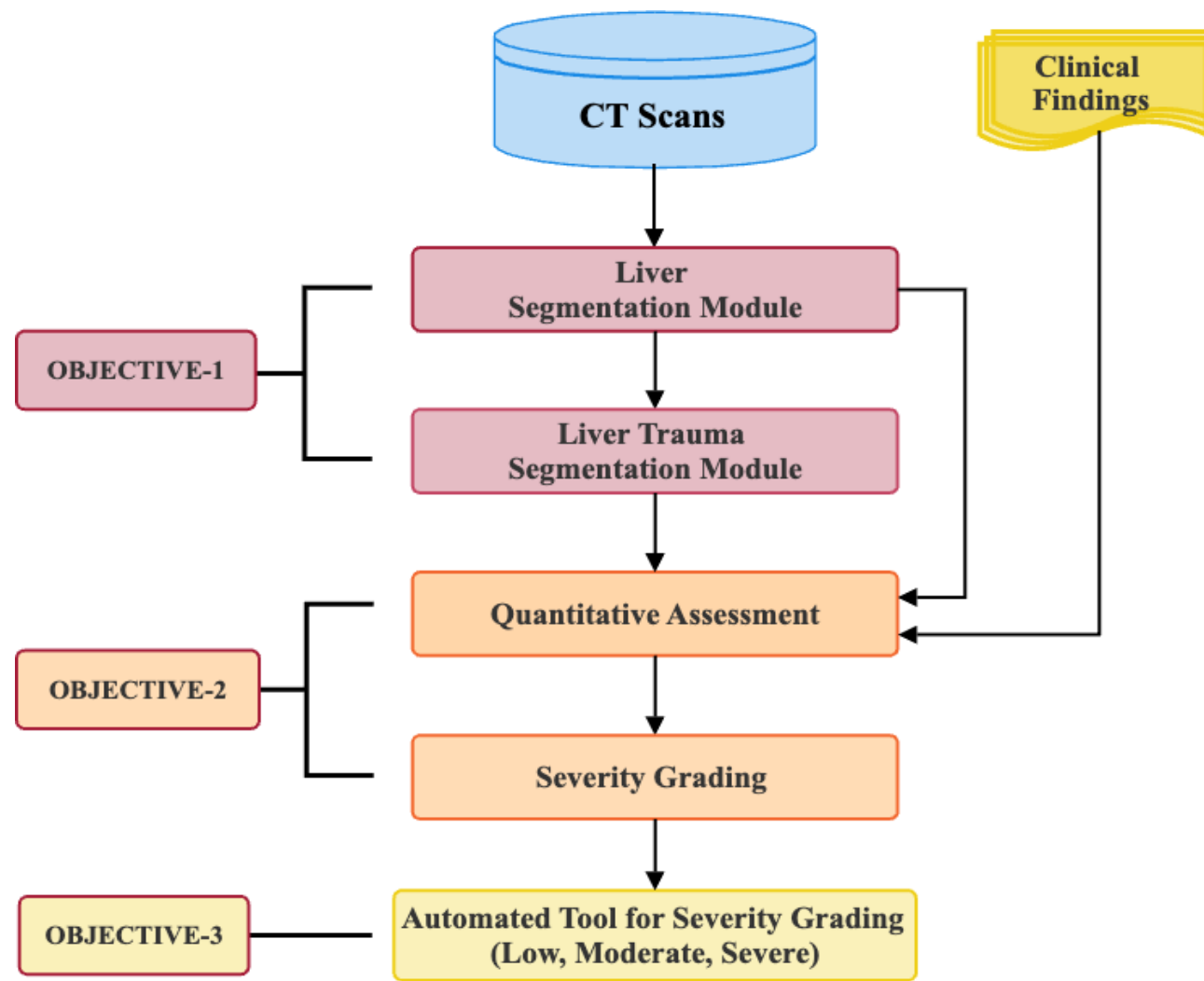


Figure 5.1: Decision Support System for Liver Trauma Assessment workflow

Course Work Details

S. No.	Title of Course Work	Credits	Status
1	Advanced Research Methodology (ARM)	4	To be Taken (TBT)
2	Research and Publication Ethics (RPE)	2	In Progress (IP)
3	Medical Image Processing (Customized)	3	DAC approval pending
4	Deep Learning and its Applications	4	DAC approval pending



Time Plan

Events	1 st semester	2 nd semester	3 rd semester	4 th semester	5 th semester	6 th semester	7 th semester
Registration for mandatory course work	Immediately after the registration						
Constitution of DAC	Within 1 month						
PreDAC meeting	Within 2 month						
IPAC presentation deadline	02-11-2024						
Registration Renewal deadlines	02-11-2024	02-05-2025	02-11-2025	02-05-2026	02-11-2026	02-05-2027	
DAC presentation deadlines		DAC1 02-05-2025	DAC2 02-11-2025	DAC3 02-05-2026	DAC4 02-11-2026	DAC5 02-05-2027	DAC6 02-11-2027
Course Work completion (12 credits)							
1 st draft of review/ original article submission to guide							
Review/ original article submission to journal							
2 nd article preparation and submission							
Conference presentations	As approved by the guide						
Synopsis presentation							
Submission of NOC request to CDS for thesis submission	Within 3 months of synopsis presentation						
Thesis submission to Registrar Evaluation	Immediately after NOC						

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References II

- [10]F. Coccolini *et al.*, “Liver trauma: WSES 2020 guidelines,” *World Journal of Emergency Surgery*, vol. 15, no. 1, Mar. 2020, doi: <https://doi.org/10.1186/s13017-020-00302-7>.
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Thank You